

THAMSANQA NDLAKUSE

IT Graduate | Systems Architect, SQL, Python, Azure

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SUMMARY

IT graduate with hands-on experience in data systems and Decision Support Systems, completing BSc (Hons) in Computer Science and IT at North-West University. Built end-to-end systems for loan management and fraud monitoring using SQL Server, Azure, and Power BI. Gained enterprise IT exposure through government internship working with EMIS and SA-SAMS data infrastructure.

EDUCATION

BSc Honours in Computer Science and Information Technology, North-West University – Potchefstroom Campus

2026 – Present

Research Topic: Implementing Management Science into administrative systems

Key Modules: Data Warehouses, Databases, Information Systems Engineering, Decision Support Systems, Research

Bachelor of Science in Information Technology, North-West University – Potchefstroom Campus

2023 – 2025

GPA: 70/100

Key Modules: Artificial Intelligence, Decision Support Systems, Databases, IT Developments, Structured & Object-Oriented Programming

EXPERIENCE

ICT Intern (Vac Work), Department of Basic Education

Jun 2024 – Jul 2024

- University-based practical internship under the Government IT Officer, gaining exposure to enterprise IT systems and data infrastructure such as Education Management Information System (EMIS) and South African School Administration System (SA-SAMS)
- Developed understanding of data governance, security policies, and IT infrastructure in a government environment

SKILLS

Programming & Scripting: Python, C#, Java, JavaScript, SQL, React

Cloud & Data Infrastructure: Microsoft Azure, Azure Stream Analytics, Power BI, MSSQL Server, SSMS, PostgreSQL

Systems & Tools: Active Directory, Bash, Hyper-V

IT Methodologies: Data governance, systems analysis, database design, object-oriented programming, Agile

Soft Skills: Analytical thinking, problem-solving, communication

CORE STRENGTHS

Critical Thinking, Digital Guru, Leader by Nature, Attention to Detail, Problem Solver, Adaptability & Continuous Learning, Customer-centric Teamwork & Collaboration

PROJECTS

Loan Management System

2024 (Full year)

- End-to-end system for managing loan records, borrower data, and repayment tracking, requiring rigorous data modelling, logic design, and analytical thinking
- Applied database design (SSMS), object-oriented programming (C#), and systems analysis methodologies

Internal Text Messaging Application

2025 (Half year)

- Designed and built a secure, database-backed corporate communication application, covering requirements analysis, architecture design, and implementation

Credit Card Fraud Monitoring (Stream Processing)

2026 (Half year)

- Designed and built a stream pipeline of transaction events
- Used Microsoft Azure to facilitate stream analytics and Power BI for data visualisation

Research Artefact Development

2026 (Full Year)

- Investigating how prescriptive analytics can be implemented within Learning Management Systems to support the monitoring and improvement of student academic performance
- Delivering a decision support system driven by prescriptive techniques to monitor and promote students for academic success

Project Manager: Demi Application and Claims Management System

2026 (Half year)

- Currently managing a web development project for automating the lifecycle of appointing student demonstrators and managing their claims for a North-West University department

Microservices: Requests Support System

2026 (Half year)

- A microservices-based support request system for a company, designed to handle user support requests efficiently
- Implementing .NET Core, Ocelot API Gateway, Consul, and Serilog for real-time data processing

KEY ACHIEVEMENTS

Student Assistant, CMPG212 (C# Programming)

- Assisting second years in onboarding C# Advanced interface programming and invigilating exam sessions

Facilitator, CMPG122 (Interface Programming)

- Creating and facilitating lecture sessions and reiterating and clarifying content to ensure a first-year programming beginner's on-ramp
- Managing all assistants